Multi-Step Linear Equations

Concept

A multi-step linear equation is solved by preserving equality at each step.

Plan

Expand → collect terms → isolate the variable → check.

Fractions

Multiply every term on both sides by the LCD.

No solution

A contradiction such as $0 = 5$.

All values

An identity such as $0 = 0$, on the allowed domain.

Example 1

Solve $3(x - 2) = 2x + 5$.

Solution

1. Expand: $3x - 6 = 2x + 5$.
2. Subtract $2x$ and add 6: $x = 11$.
3. Check: $3(11 - 2) = 27 = 2(11) + 5$.

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Example 2

Solve $x/3 - 1/2 = x/6 + 1$.

Solution

1. Multiply every term by 6: $2x - 3 = x + 6$.
2. Collect terms: $x = 9$.
3. Check: $3 - 0.5 = 1.5 + 1 = 2.5$.

Multi-Step Linear Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $4(2x - 1) = 3x + 16$.

2. Solve $(x + 1)/4 = (x - 2)/3$.

3. Solve $5x - 7 = 5x + 2$.

4. Solve $3(x + 2) = 3x + 6$.

5. Two consecutive integers total 47. Find them and verify.

Multi-Step Linear Equations

Practice answers | check after trying

1. $8x - 4 = 3x + 16$ gives $5x = 20$, so $x = 4$.

2. $3x + 3 = 4x - 8$ gives $x = 11$.

3. No solution: $-7 = 2$ is false.

4. All real x: both sides are identical.

5. $n + (n + 1) = 47$ gives $n = 23$. The integers are 23 and 24, and their sum is 47.